Reproducing and Extending Confident Learning for Detecting Label Errors in Classification Data

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Abstract

Confident learning (CL) estimates and ranks label errors using out-of-sample class probabilities. This project first reproduces a released CIFAR-10 benchmark from Northcutt et al. and then tests how well the method generalises beyond the paper's synthetic benchmark. Running the original `cleanlab` v0.1.0 on the authors' released probabilities regenerated all five released label-error masks bit-for-bit in the main setting, reproducing the paper's confident-joint result exactly (precision 0.771, recall 0.836, F_1 0.802); 45 of 60 masks across all twelve noise/sparsity settings were identical and the rest differed on at most 9 of 50,000 examples. The paper's joint-estimation RMSE was also reproduced: all 24 values in its Table S1 were matched after rounding. Current `cleanlab` 2.9.0 yields $F_1 = 0.790$ with 98.5% mask agreement, a measurable software-version effect in the filtering step. Extensions used real human annotations from CIFAR-10N and CIFAR-100N, and a new controlled text dataset, AGNews-CLNoise. On the full 50,000-image CIFAR-10N aggregate labels, CL achieved $F_1 = 0.546$; on a balanced 10,000-image CIFAR-100N subset it achieved $F_1 = 0.650$. On AGNews-CLNoise, the original 20% cyclic setting achieved $F_1 = 0.800$, while a 24-run sensitivity study showed that performance depended strongly on the structure of corruption and the classifier supplying predicted probabilities. A blind human review and a pinned, order-reversed Qwen judge further showed that semantic review can explain apparent false positives, but that an LLM judge is position-sensitive. All scripts, derived data, source revisions, results, and deterministic hashes are released in a public repository.

1 Introduction and Paper Justification

Supervised classifiers normally treat dataset labels as ground truth, even though labels may be wrong because of annotation mistakes, ambiguous examples, automatic collection, or overlapping class definitions. A label-error detector therefore has two related goals: estimate how much noise exists and identify which individual examples should be reviewed.

Northcutt et al. propose CONFIDENT LEARNING, a model-agnostic framework that estimates the joint distribution between observed noisy labels and latent true labels using out-of-sample predicted probabilities (Northcutt et al., 2021b). CL uses class-specific confidence thresholds to count confident disagreements, constructs a *confident joint*, and then ranks or prunes suspected label issues. The paper makes three claims relevant to this project: (1) under stated conditions CL consistently estimates the noisy-true label joint and can exactly identify errors; (2) it is not tied to one classifier or modality; and (3) removing suspected errors can improve learning with noisy labels. The original evaluation combines controlled CIFAR-10 corruption with qualitative and downstream studies on images and text.

The paper is a strong basis for a novel replication project on several independent criteria. *Venue quality:* it appeared in the refereed, diamond-open-access *Journal of Artificial Intelligence Research* (JAIR), which the CORE2020 journal ranking (CORE's final journal list) rates **A**.[1] *Impact:* Google Scholar listed 1,557 citations as of 8 November 2026, and the method underlies the widely used open-source `cleanlab` library. *Reproducibility:* the authors released code, the exact noisy labels, out-of-sample probabilities, and every method's output mask, so each stage of the result can be checked. *Relevance:* its modality-independent claim creates a meaningful extension question: does the procedure retain useful error-detection performance under real human noise and

[1] Paper: https://www.jair.org/index.php/jair/article/view/12125; CORE ranking: https://portal.core.edu.au/jnl-ranks/145/.

under a new text construction?

This report addresses four research questions:

1. Can the released CIFAR-10 label-error results be reproduced from the authors' artifacts, and how does the current software behave on them?
2. How does CL perform on real human annotations in CIFAR-10N and CIFAR-100N?
3. How stable is CL on a controlled text dataset when noise rate, noise structure, random seed, and classifier are varied?
4. Can blind human review and a reproducible LLM judge clarify disputed labels without being treated as infallible ground truth?

2 Original Work and Data

2.1 Method and evaluation framework

Let $\tilde{y}$ denote the observed label and y^* the unknown true label. Given out-of-sample class probabilities $\hat{p}(y = j \mid x)$, CL calculates a confidence threshold for each observed class and counts examples that confidently support a different class. These counts are calibrated to estimate $P(\tilde{y} = i, y^* = j)$. Entries away from the diagonal correspond to possible label transitions. Crucially, cross-validation is used so that an example is not judged by a model trained on that same example.

The original paper evaluates three questions. First, on synthetic CIFAR-10 noise, root mean squared error measures estimation of the known noisy-true joint. Second, individual error detection is evaluated by precision, recall, and F_1 (Sokolova and Lapalme, 2009):

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \tag{1}$$

Third, classifiers are retrained after suspected errors are removed and evaluated by held-out accuracy. The original CIFAR-10 study varies label-noise rate and transition-matrix sparsity and compares five CL variants with several noisy-label baselines. The most computationally expensive downstream results average ten ResNet-50 trials; those full training trials were outside the CPU budget of this project.

On its own benchmark the method was highly successful. For joint estimation at 20% and 40% noise, CL averaged an RMSE of 0.004 against the true joint, versus 0.006 for the simpler confusion-matrix estimate. For label-error detection on CIFAR-10, the authors' released confident-joint masks give F_1 between 0.69 and 0.84 across the twelve released noise/sparsity settings, including $P = 0.771$, $R = 0.836$, $F_1 = 0.8025$ at 40% noise and 60% sparsity, the values reported in the paper's Table 4. For downstream learning (the paper's Table 2), training on CL-cleaned data at 40% noise and 60% sparsity reached 87.2% CIFAR-10 test accuracy for the best variants (PBNR and C+NR) and 86.9% for the confident joint, compared with 73.6% for INCV, the strongest non-CL baseline, and 57.3% for training on the noisy labels directly.

The extensions additionally report Matthews correlation coefficient (MCC), which uses all four confusion-matrix cells and is useful when errors are rare (Matthews, 1975), and average precision (AP), which evaluates the ranking induced by $1-$label-quality score. "Lift" is reported only as precision divided by error prevalence: it measures how concentrated true errors are in the flagged set relative to random selection, not ranking quality.

2.2 Datasets in the original paper

Table 1 summarises every dataset used in the original paper. CIFAR-10 is the main quantitative error-detection benchmark because its synthetic corruption provides known ground truth. The other datasets test different scales, modalities, or naturally occurring label problems.

CIFAR-10 itself is distributed as binary batches of 32×32 colour images with integer class labels; the authors' artifact stores the training images as PNG files in one folder per class.[2] The authors' released replication artifact[3] wraps it in a linked set of files for each of the twelve noise/sparsity settings: (i) a JSON mapping from each training-image path to its noisy class, where the path's folder name encodes the clean class; (ii) a $50{,}000 \times 10$ float16 matrix of four-fold out-of-sample ResNet-50 probabilities; and (iii) for each of five methods, a 50,000-element Boolean keep mask. Table 2 shows genuine rows from the 40%-noise, 60%-sparsity setting, one for each outcome type.

[2] CIFAR-10: https://www.cs.toronto.edu/~kriz/cifar.html.

[3] Authors' artifact repository (commit 2f31556): https://github.com/cgnorthcutt/confidentlearning-reproduce.

Dataset	Scale / format	Role in the paper	Important preparation
CIFAR-10 (Krizhevsky and Hinton, 2009)	60,000 32×32 RGB images, 10 classes	Joint estimation, error detection, and downstream learning	The 50,000 training labels are corrupted with asymmetric class-conditional matrices; the 10,000 test labels remain unchanged.
ImageNet (Russakovsky et al., 2015)	1.28M training and 50,000 validation images, 1,000 classes	Real label/ontology problems and filtered retraining	ResNet-18/50 predictions are used; targeted cleaning is compared with random removal.
MNIST (LeCun et al., 1998)	70,000 greyscale digit images, 10 classes	Errors in a dataset often assumed clean	A small CNN produces cross-validated probabilities; examples are ranked by self-confidence.
WebVision (Li et al., 2017)	More than 1M web images, 1,000 classes	Large-scale qualitative error discovery	A pretrained ResNeXt supplies probabilities; CL ranks likely errors without retraining on WebVision.
Amazon Reviews (He and McAuley, 2016)	Approximately 10M filtered reviews, 3 sentiment classes	Text, class imbalance, and non-deep classification	Two- and four-star reviews are removed; five-fold multinomial logistic regression uses fastText representations.

Table 1: Original datasets and their distinct evaluation roles.

Row	Clean	Noisy	$\hat{p}_{\text{noisy}}$	$\hat{p}_{\text{clean}}$	$\max \hat{p}$	Kept?
1	airplane	automobile	.068	.642	.642	no (TP)
0	airplane	airplane	.208	.208	.473	no (FP)
30	airplane	automobile	.205	.131	.458	yes (FN)
18	airplane	airplane	.355	.355	.355	yes (TN)

Table 2: Genuine rows of the released CIFAR-10 artifact (40% noise, 60% sparsity). "Row" is the array index; e.g. row 1 is image `airplane/10010_airplane.png` with JSON noisy label 1 (automobile). $\hat{p}$ values are ResNet-50 probabilities; "Kept?" is the confident-joint keep mask, with the outcome relative to the clean label.

3 Replication of the Original Work

3.1 Scope and environment

The replication has four stages, all on the authors' released data at artifact commit `2f31556`; scripts, outputs, and hashes are in the public project repository.[4]

Stage 1 (evaluate released outputs). For the 40%-noise, 60%-sparsity setting (50,000 images; 19,981 actual corruptions, 39.96%), the script reconstructs clean labels from image paths, inverts each released keep mask to a flagged-error mask, and recomputes TP, FP, FN, P, R, and F_1.

Stage 2 (run the original code). The artifact README states that the paper used `cleanlab` v0.1.0. That version predates current NumPy, so it was run in an isolated Python 3.11.18 environment (NumPy 1.23.5, scikit-learn 1.2.2, SciPy 1.10.1). The script makes exactly the calls in the authors' work log (`compute_confident_joint` and `get_noise_indices` with each pruning method) on the unmodified float16 probabilities, for all twelve released settings, and compares every regenerated mask with the released one.

Stage 3 (current software). The five methods were re-run with `cleanlab` 2.9.0 (Python 3.12.1, NumPy 2.5.2, scikit-learn 1.9.0, GitHub Codespace). Because the float16 rows sum to 0.9868-1.0000 and the current library validates its inputs, they were converted to float64 and renormalised for this stage only.

Stage 4 (joint estimation). Following the authors' work log, the true joint is the released noise matrix multiplied by the clean-label prior, $Q_{\tilde{y},y^*} = P(\tilde{y} \mid y^*)\,P(y^*)$; $\hat{Q}$ is obtained from `estimate_joint` using either the confident joint or the argmax confusion matrix ($C_{\text{confusion}}$), and RMSE $= \left(\frac{1}{m^2}\sum_{i,j}(\hat{Q}_{ij} - Q_{ij})^2\right)^{1/2}$. This was run in both environments for all twelve settings.

The scope is deliberately the joint-estimation and label-error detection results, not the paper's ten-trial ResNet-50 retraining benchmark, which exceeds the CPU budget. Reusing the released out-of-sample probabilities isolates CL's counting and filtering behaviour, which is the paper's methodological contribution.

3.2 Replication results

Results and closeness. For the released confident-joint mask, $TP = 16{,}710$, $FP = 4{,}955$, and $FN = 3{,}271$, giving $P = 0.7713$, $R = 0.8363$, and $F_1 = 0.8025$, matching the paper's Table 4 (77%, 84%, 80%). The paper's Table 4 reports four settings; Table 4 compares all

[4] https://github.com/marufkhan09/comp8240-confident-learning. All results and scripts in this report correspond to repository commit `8577ccc`.

Method	Paper	Released	v0.1.0	v2.9.0	Agree
$C_{\text{confusion}}$	.79	.7880	.7880	.7879	99.98%
$C_{\tilde{y},y^*}$	.80	**.8025**	**.8025**	.7900	98.51%
PBC	.79	.7946	.7946	.7945	99.96%
PBNR	.80	.8004	.8004	.8002	99.97%
C+NR	.78	.7830	.7830	.7829	99.95%

Table 3: F_1 at 40% noise, 60% sparsity: the paper's Table 4 (two decimals, as printed), the authors' released masks, masks regenerated with the original `cleanlab` v0.1.0 (all five identical to the released masks), and current `cleanlab` 2.9.0 with its example-wise agreement with the released mask. $C_{\text{confusion}}$ is the argmax-disagreement method; $C_{\tilde{y},y^*}$ is the confident joint; C+NR is "both".

of them, and every reported F_1 is matched after rounding. The comparison is restricted to these four noise-sparsity combinations; the remaining eight released configurations (Table 5) are reported separately and are not presented as replications of a paper table. Running the original code reproduced the method itself, not only its outputs: with `cleanlab` v0.1.0, all five masks in the main setting were bit-for-bit identical to the released masks (Table 3). Across all twelve settings, 45 of the 60 regenerated masks were identical, including the confident-joint mask in every setting (Table 5). The other 15 (seven CL-PBNR, seven CL-both, and one argmax mask) differed from the released mask on at most 9 of 50,000 examples, changing F_1 by at most 0.0002. These residual differences were deterministic across our runs, so they are most plausibly tie-breaking effects of the original computing environment rather than a methodological discrepancy; they were not investigated further.

Why current software differs. Under `cleanlab` 2.9.0, four methods are essentially unchanged, but the confident-joint mask differs from the released mask on 746 examples (98.51% agreement) and its F_1 falls to 0.7900. The two runs did not receive literally identical inputs, because only the modern path required float64 renormalisation. This was tested directly: running `cleanlab` v0.1.0 on the same renormalised probabilities changes only 18 of 50,000 mask decisions and reduces F_1 only from 0.8025 to 0.8024. Renormalisation therefore explains a negligible part of the discrepancy: even if every one of those 18 changes overlapped with `cleanlab` 2.9.0's, at least 728 of its 746 differences remain. Most of the gap is associated with changes in the modern `cleanlab` implementation and its software environment, not with the data, the released masks, the metric calculation, or an error in the paper. Reporting all three columns separates repeatability of the artifact, reproducibility of the method, and robustness to software change.

Joint estimation. Table 6 reproduces the paper's first evaluation question. All 24 RMSE values match the paper's Table S1 after rounding to three decimals. Averaged over the eight 20% and 40% settings, the confident joint gives 0.0042 and $C_{\text{confusion}}$ gives 0.0055, matching the paper's summary values of 0.004 and 0.006. The advantage of the confident joint narrows at 70% noise, as in the paper. Unlike the error masks, the estimated joint is essentially unchanged under `cleanlab` 2.9.0 (maximum difference 9×10^{-6}). This suggests that the 746-example mask discrepancy above arises in the modern filtering step rather than in counting or calibrating the confident joint.

All four stages produced identical outputs when re-run. The repository README's Quickstart lists the command, downloads, and approximate CPU time for every experiment in this report, and each script saves its output hashes.

4 Construction of New Data

The extension was designed around two complementary forms of evidence. CIFAR-N provides *existing* real human annotation noise; AGNews-CLNoise is a *newly constructed* text benchmark with exactly known injected errors. In every experiment, the model produces out-of-sample probabilities, CL flags issues, and the flag is compared with a reference error mask. Table 7 summarises the construction pipeline step by step; the subsections explain the choices.

4.1 CIFAR-10N and CIFAR-100N

CIFAR-N adds Mechanical Turk labels to the original CIFAR training images (Wei et al., 2022). The official annotation repository was pinned at commit `49df7d8`.[5] The image ordering was verified by checking that its `clean_label` array exactly matched the official torchvision targets. This is important: a silent order mismatch would make every subsequent metric meaningless.

[5] CIFAR-N labels: https://github.com/UCSC-REAL/cifar-10-100n.

Method	20% / 0.0 paper | ours	20% / 0.6 paper | ours	40% / 0.0 paper | ours	40% / 0.6 paper | ours
$C_{\text{confusion}}$	71 | 70.9	72 | 72.4	84 | 84.0	79 | 78.8
$C_{\tilde{y},y^*}$	75 | 75.0	78 | 77.7	84 | 83.6	80 | 80.2
PBC	76 | 76.4	76 | 76.1	84 | 84.4	79 | 79.5
PBNR	77 | 76.7	79 | 78.7	85 | 85.0	80 | 80.0
C+NR	78 | 78.1	78 | 78.1	84 | 84.2	78 | 78.3

Table 4: Every setting in the paper's Table 4 (noise / sparsity): reported F_1 (%, as printed) versus F_1 of the released masks, which `cleanlab` v0.1.0 regenerates. All 20 F_1 values agree after rounding. Across all 60 reported precision, recall, and F_1 values, 59 round to the printed value. The sole exception is PBC precision at 20% noise and 0.0 sparsity: 0.6349 from the released artifact versus 0.64 in the paper, a difference of 0.0051.

Noise	Sparsity			
	0.0	0.2	0.4	0.6
20%	.750	.753	.787	.777
40%	.836	.828	.842	.803
70%	.828	.836	.782	.689

Table 5: Confident-joint F_1 in all twelve released settings. Masks regenerated with `cleanlab` v0.1.0 were identical to the released masks in every setting.

		Sparsity			
Noise	Count	0.0	0.2	0.4	0.6
20%	$C_{\tilde{y},y^*}$	.0042	.0041	.0036	.0037
	$C_{\text{confusion}}$	.0058	.0056	.0049	.0055
40%	$C_{\tilde{y},y^*}$	.0040	.0043	.0041	.0052
	$C_{\text{confusion}}$	.0049	.0051	.0055	.0070
70%	$C_{\tilde{y},y^*}$	.0108	.0102	.0146	.0174
	$C_{\text{confusion}}$	.0112	.0111	.0151	.0192

Table 6: RMSE of the estimated joint $\hat{Q}$ (`cleanlab` v0.1.0, released probabilities). All 24 values round to the paper's Appendix Table S1; `cleanlab` 2.9.0 agrees to within 10^{-5}.

For the CIFAR-10N pilot, a deterministic stratified subset of 5,000 images was selected with seed 42. Images were resized to 96×96, normalised with ImageNet statistics, and passed through a frozen ImageNet-pretrained ResNet-18. The resulting 512-dimensional features were cached. A standardised logistic regression model then generated four-fold out-of-sample probabilities. The same cached images and the fold assignment derived from aggregate labels were reused for all five human-label sets: aggregate, worst-case, and three individual annotators. The full aggregate-label experiment used all 50,000 images and the same representation; caching the features makes repeat runs fast on CPU.

For CIFAR-100N, a 10,000-image subset selected exactly 100 images from each of 100 fine-grained classes. The official `noisy_label` was compared with `clean_label`; coarse labels were excluded because they define a different 20-class task. The same frozen ResNet-18 and logistic-regression pipeline was used, making the CIFAR-10N/CIFAR-100N comparison controlled with respect to feature extraction and classifier family.

4.2 AGNews-CLNoise

AGNews-CLNoise was derived from the 120,000-example AG News training split (Zhang et al., 2015), pinned to Hugging Face revision `eb185aad`; the complete revision is recorded in the repository.[6] A stratified sample of 10,000 articles contains 2,500 examples from each of World, Sports, Business, and Sci/Tech. The source label is treated as the clean reference, with the limitation that the original dataset may itself contain ambiguity.

The baseline changes exactly 500 labels per class (2,000 total, or 20%) using the fixed cycle World→Sports→Business→ Sci/Tech→World. The follow-up sensitivity study keeps the same articles and four folds while varying four noise rates (10-40%), cyclic versus symmetric corruption, and seeds 42-44. This produces 24 controlled runs. Symmetric corruption randomly distributes each selected error over the other three classes; cyclic corruption concentrates it in one competing class. Each run generates four-fold probabilities with a TF-IDF (unigrams and bigrams, 30,000-feature maximum) logistic-regression pipeline. An argmax disagreement rule is included as a simple baseline.

Because the AG News dataset card listed its licence as unknown during the project, raw text was downloaded only into the ignored `data/raw` directory. The public manifest stores source indices, labels, and SHA-256 text hashes, allowing integrity

[6] AG News: https://huggingface.co/datasets/fancyzhx/ag_news.

Step	CIFAR-10N	CIFAR-100N	AGNews-CLNoise
1. Source	CIFAR-10 train images + CIFAR-N human labels (repo commit `49df7d8`)	CIFAR-100 train images + CIFAR-N labels (same commit)	AG News train split, 120,000 articles (HF revision `eb185aad`)
2. Alignment check	CIFAR-N `clean_label` must equal torchvision targets in order	Same check against CIFAR-100 fine labels	Text hashed (SHA-256); manifest stores index, labels, hash
3. Sampling	Pilot: 5,000 images, stratified on aggregate labels, seed 42; full run: all 50,000	10,000 images, stratified on clean labels (100 per class), seed 42	10,000 articles, stratified (2,500 per class), seed 42
4. Noisy labels	Real: aggregate, worst-case, and three individual annotator label sets	Real: official `noisy_label` (coarse labels excluded)	Injected: exactly $r \times 2{,}500$ per class at $r \in \{10, 20, 30, 40\}\%$, cyclic or symmetric, seeds 42-44
5. Representation	Resize to 96×96, ImageNet normalisation, frozen ResNet-18 → 512-d features (cached)	Same as CIFAR-10N	TF-IDF, unigrams + bigrams, 30,000 features, sublinear TF
6. Probabilities	Standardised logistic regression, stratified 4-fold out-of-sample, seed 42	Same as CIFAR-10N	Logistic regression, the same 4 stratified folds for every run
7. Reference errors	Noisy label ≠ CIFAR-N clean label	Noisy label ≠ clean label	Exactly the injected positions

Table 7: Construction pipeline for the new datasets. Each row is one processing step; every random choice is seeded and recorded in the repository.

checks without republishing the articles. The examples in Table 8 illustrate the committed formats.

4.3 Blind annotation and LLM-judge protocol

Thirty cases were sampled from CL flags that were *not* deliberately corrupted in the verified 20%-cyclic AG News run. Under the synthetic reference these are apparent false positives. Sampling was approximately balanced across the four source classes. In a blind review, the human annotator saw the article and two randomly ordered candidates: the original AG News label and the model's alternative. Their roles and earlier model decisions were hidden. The choices were candidate A, candidate B, ambiguous, or neither. A three-category mapping (original, alternative, ambiguous/neither) and agreement statistics were written before the review. This is narrower than the proposal's plan of about 100 cases stratified across TP, FP, FN, and TN. With a single annotator, the review effort was concentrated on apparent false positives, the only outcome whose correctness the synthetic reference cannot settle; true positives and false negatives are decided by construction.

Because only one human annotator was available, an independently executed, pinned, order-controlled LLM judge was used as a supplementary annotator. This follows the unit's permitted LLM-as-judge approach but does not replace a human-human reliability study. The judge was Qwen2.5-0.5B-Instruct (Qwen Team, 2024), pinned to revision `7ae5576`[7] and run locally. It replaced the proposal's tentative Prometheus 2 (Kim et al., 2024), whose 7B+ models were impractical on the available CPU-only Codespace; local execution also kept article text away from commercial APIs. The small judge was a deliberate CPU-compatible trade-off and may have contributed to the position sensitivity reported below; this cannot be isolated without comparison to a larger judge. Each case was judged twice: pass 1 used a deterministically random candidate order and pass 2 reversed it. The prompt defined the four topics, hid label provenance, required A/B/AMBIGUOUS/NEITHER, and requested one-sentence justification. Greedy decoding, the exact prompt, model revision, seeds, and software versions were recorded. Raw text and raw responses stayed private; the public output contains identifiers, hashes, candidate roles, parsed choices, and consistency indicators. Cohen's κ measures agreement beyond chance (Cohen, 1960).

5 Results on New Data

5.1 Relation to the replication

The replicated benchmark and the extensions differ in three deliberate ways, so their numbers are related but not identical quantities. (1) *Noise source:* the replication's errors are synthetic class-conditional corruptions with exact ground truth; CIFAR-N errors are real human disagreements with

[7] Model: `https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct`.

Source	Index	Clean	Noisy	Pred.	Flag	Score	Other stored metadata
CIFAR-10N	19	6	6	0	yes	.0005	actual-error Boolean; original image index
CIFAR-100N	1	29	63	29	yes	1.9×10^{-8}	actual-error Boolean; original image index
AGNews	14	Business	Sci/Tech	Business	yes	.1561	source index; text hash; run/seed
AGNews	38	Business	Sci/Tech	Sci/Tech	no	.8459	source index; text hash; run/seed

Table 8: Snippets of the row-oriented derived data. "Score" is label quality; lower values indicate more likely errors. Text and images remain in their source datasets rather than being republished.

a reference label that may itself be imperfect; AGNews errors are synthetic again but in text. (2) *Probability model:* the replication uses the authors' four-fold ResNet-50 trained on the noisy labels; the image extensions use a frozen ImageNet ResNet-18 with four-fold logistic regression, and AGNews uses TF-IDF with four-fold logistic regression. (3) *Metrics:* the replication reports the paper's P, R, and F_1; the extensions add MCC and AP because error prevalence varies from 9% to 40% across datasets. Holding the CL step fixed while changing the noise and the model is what allows the comparison in Table 9 to test the paper's generality claim.

At a matched 20% noise rate, the replicated CIFAR-10 confident-joint F_1 ranges from 0.750 to 0.787 across sparsities (Table 5), and the constructed AGNews benchmark with 20% cyclic noise gives a comparable $F_1 = 0.800$. This supports the claim that CL transfers across modalities when the noise is synthetic and the classifier is adequate. At 40% the picture splits: CIFAR-10 stays at 0.803-0.842, AGNews symmetric noise is similar (0.847), but AGNews cyclic noise falls to 0.664 (Section 5.3). Real human noise is harder: CIFAR-10N and CIFAR-100N reach F_1 of 0.546 and 0.650, mainly through lower precision.

5.2 Real human annotation noise

Table 9 shows that the strong synthetic result does not transfer unchanged to real human annotations. In the 5,000-image CIFAR-10N pilot, CL recovered 355 of 466 disagreements but also flagged 1,076 cases that matched the clean reference, producing high recall (0.762) and low precision (0.248). On all 50,000 aggregate labels, precision and F_1 rose to 0.482 and 0.546 while recall fell to 0.629. Because this is one fixed pilot subset rather than a randomised sample-size experiment, the difference is an association, not a causal estimate of the benefit of more data.

Across the five CIFAR-10N label sets, F_1 ranged from 0.374 (aggregate, 9.3% disagreement prevalence) to 0.664 (worst-case, 40.5% prevalence), while MCC stayed in the narrower 0.337-0.390 range. Thus, the higher F_1 for the worst-case labels partly reflects that a flagged example is more likely to be wrong when errors are common; it does not by itself show easier intrinsic detection. CIFAR-100N had high recall (0.845) but modest MCC (0.356), suggesting that CL found most disagreements at the cost of many flags. The much lower clean-label classification accuracy (0.364) also makes this 100-class task harder for the frozen-feature classifier.

5.3 Controlled text sensitivity study

The 20%-cyclic AG News baseline recovered 1,701 of 2,000 injected errors and exactly reproduced the original pilot counts. Across all 24 runs, CL had higher mean F_1 than argmax disagreement in seven of eight aggregate configurations (Figure 1), although at 40% noise both differences are within seed-to-seed variation (cyclic 0.6636 versus 0.6611; symmetric 0.8468 versus 0.8480). The largest advantage occurs at 10% noise, where errors are rare.

The most informative failure is 40% cyclic corruption: CL's mean F_1 falls to 0.6636, compared with 0.8468 under symmetric noise. A plausible explanation is that cyclic noise concentrates all changed labels from one class into the same alternative, creating a strong competing mapping that the classifier can learn; symmetric noise spreads those changes across three alternatives. This is an interpretation of the controlled observation, not a general causal claim about all text datasets. It also shows why reporting only one synthetic noise mechanism would overstate generality.

Classifier choice mattered even when articles, folds, rates, structures, and seeds were fixed. At 20% cyclic/symmetric noise, mean CL F_1 was

Dataset / scope	n	Errors	Flags	P	R	F_1	MCC	AP	Clean acc.
Replication: CIFAR-10 synthetic	50,000	19,981	21,665	.771	.836	.802	-	-	-
CIFAR-10N aggregate pilot	5,000	466	1,431	.248	.762	.374	.337	.427	.692
CIFAR-10N aggregate full	50,000	4,505	5,878	.482	.629	.546	.499	.647	.820
CIFAR-100N fine pilot	10,000	3,954	6,329	.528	.845	.650	.356	.692	.364
AGNews 20% cyclic	10,000	2,000	2,251	.756	.851	.800	.749	.730	.856

Table 9: Main extension results, with the replicated confident-joint result (40% noise, 60% sparsity) as the reference row. For CIFAR-N, "errors" are disagreements between human noisy labels and released clean references; for CIFAR-10 and AGNews they are injected corruptions. MCC, AP, and accuracy are not available for the released masks, which store no scores.

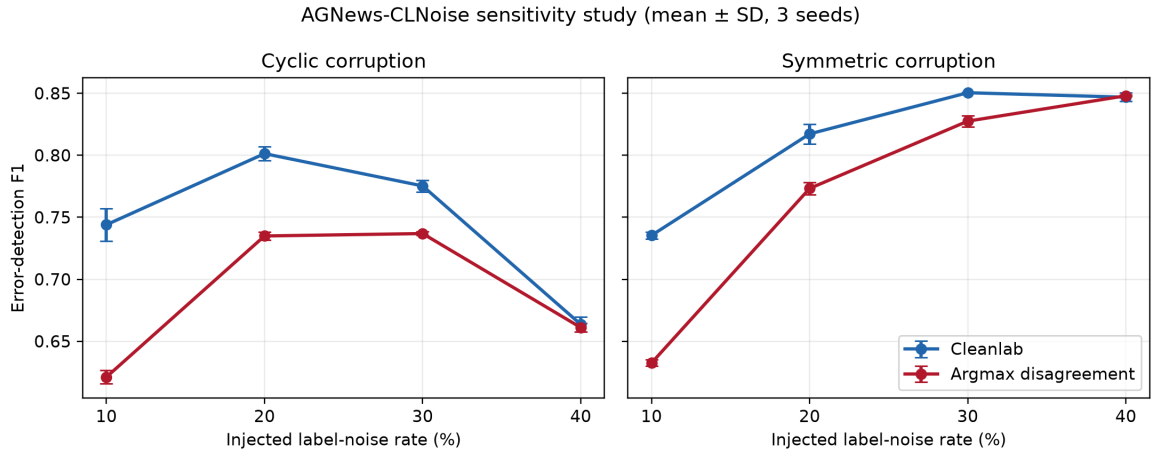


Figure 1: AGNews-CLNoise sensitivity study. Points are means and error bars are standard deviations over corruption seeds 42-44. Articles and folds are held fixed.

0.801/0.817 for logistic regression, 0.838/0.834 for Complement Naive Bayes, and 0.777/0.802 for calibrated linear SVM. Complement Naive Bayes was best in this experiment, but the broader conclusion is more cautious: CL is model-agnostic as an algorithm, yet its empirical quality depends on the probabilities supplied by the underlying classifier.

5.4 Human annotation and LLM-as-judge

The blind human reviewer selected the original label in 15 cases and the model-supported alternative in 15; no case was assigned ambiguous/neither. This result indicates that some apparent false positives under the synthetic reference may be genuine semantic problems in the original AG News labels. However, one annotator and 30 deliberately disputed cases cannot estimate the dataset-wide error rate.

All 60 formal Qwen responses were parseable, but only 14 of 30 cases (46.7%) received the same semantic decision after candidate order was reversed. Table 10 therefore gives two distinct messages. First, agreement with the human judgement is positive and, on the order-consistent subset,

Qwen comparison	n	Agree	κ
Pass 1 vs human	30	76.7%	.533
Pass 2 vs human	30	63.3%	.313
Consistent subset vs human	14	92.9%	.837

Table 10: Formal Qwen judge agreement with the blind human review. The consistent subset is selected and is not an overall accuracy estimate.

strong. Second, the large pass difference and low order consistency reveal position sensitivity. The 92.9% consensus agreement is calculated only after selecting stable cases and must not be presented as judge accuracy on all 30. The judge is useful for prioritising or explaining cases, but the results do not support replacing independent human inspection.

As an additional external check, the released Mechanical Turk records from a follow-up CL study (Northcutt et al., 2021a) were re-aggregated for seven single-label test sets. All published counts were reproduced exactly: 4,835 human-validated errors among 11,813 CL-flagged candidates. These percentages are conditional on prior CL selection

and are not overall dataset error rates. The analysis is supplementary because it reproduces a follow-up paper rather than the central 2021 JAIR experiment.

6 Ethical Considerations

The Australian Code for the Responsible Conduct of Research emphasises honesty, rigour, transparency, fairness, respect, and accountability (National Health and Medical Research Council et al., 2018). These principles affect both the original work and this extension.

Issues in the original paper. The paper's main benchmark inherits the provenance of its data. CIFAR-10 was drawn from the 80 Million Tiny Images collection (Torralba et al., 2008), which its creators withdrew in 2020 after an audit found derogatory class labels and offensive images (Birhane and Prabhu, 2021); CIFAR-10's ten classes are benign, but its images were collected from web search without consent. ImageNet, used for the paper's real-noise study, has documented problems in its person categories that required later filtering (Yang et al., 2020). WebVision is scraped from the web and Amazon Reviews consists of user-generated text, raising copyright and consent questions about reusing content for purposes its authors never anticipated. The paper also frames removal of suspected errors as beneficial; if applied to data about people, automatic pruning could silently delete legitimate but atypical examples and shift error rates onto under-represented groups. On the positive side, the paper is open access and releases code and intermediate artifacts, which is what made this independent check possible and reflects the Code's emphasis on transparency and open dissemination.

Labels are claims, not facts. Calling an example a "label error" can conceal legitimate ambiguity, multi-label content, or a weak ontology. Automatic deletion could disproportionately remove unusual or minority examples that a classifier finds difficult. The original paper partly recognises ontology overlap, but any deployment should treat CL flags as review priorities rather than automatic proof. This report therefore uses "disagreement" for CIFAR-N and preserves an ambiguous/neither option in semantic review.

Human and LLM annotation. The blind annotation sample is small and has one human reviewer, so it cannot serve as definitive ground truth. The LLM judge may reproduce training-data bias, misunderstand short articles, and prefer the first or second option. Order reversal exposed this last risk directly. Reporting both passes, invalid-output checks, κ, and selection of the consensus subset is more honest than reporting only the most favourable agreement number. Human review should remain the final decision point for consequential relabelling.

Provenance, licensing, and privacy. Source datasets and commits are recorded, but availability does not imply permission to redistribute every item. Because the AG News licence was unclear, raw articles were not committed; public files contain source indices and text hashes. Likewise, CIFAR images remain in the source datasets and only derived predictions and metrics are committed. The LLM ran locally on the 30 cases, so article text was not sent to a commercial API. Hashes support integrity but do not make source text anonymous if an adversary already possesses the corpus; they are provenance aids, not a privacy guarantee.

Reproducibility and environmental cost. The repository records source revisions, fixed seeds, environments, commands, raw run tables, and SHA-256 hashes. Negative evidence is also retained: a full CIFAR-10N attempt with two dataloader workers failed because of Codespace shared-memory limits, after which the safe configuration used zero workers and one cross-validation job. Reporting every planned AG News configuration and recording the review mapping before annotation also reduce selective reporting and post-hoc adjustment. Frozen embeddings and caches reduce repeated CPU work. These decisions improve auditability and avoid presenting a polished result without the constraints that shaped it.

Crowd labour and use of AI. The CIFAR-N labels and the follow-up study's validations were produced by Mechanical Turk workers; reusing them should acknowledge that labour, and a publication that commissioned new labels would need fair pay and review under the *National Statement on Ethical Conduct in Human Research* (National Health and Medical Research Council et al., 2023). Consistent with the Code's transparency principle, AI use in this project is disclosed: the 30 blind decisions were made by the author alone; an earlier exploratory ChatGPT-assisted pass over the same cases is not used as evidence; the Qwen judge is

the only LLM result reported. Generative AI tools (ChatGPT/Codex and Claude) assisted with experimental planning, Python and LaTeX drafting, debugging, and language editing. The author executed the experiments, inspected the outputs, verified the numerical results and hashes, completed the blind annotations, checked the cited sources, and accepts responsibility for the final report.

7 Conclusions and Reflections

Both of the paper's CIFAR-10 estimation results were reproduced from its released artifacts. All 20 label-error F_1 values in Table 4 and all 24 joint-estimation RMSE values in Table S1 were matched after rounding. Running the original `cleanlab` v0.1.0 also regenerated the released masks themselves, so the method was reproduced and not only its outputs. Modern `cleanlab` 2.9.0 estimates the same joint but filters differently, lowering the confident-joint F_1 to 0.790. Only the ten-trial retraining experiments remained outside the CPU budget.

The generalisation results are mixed in a useful way. CL remains effective across human image annotations and synthetic text corruption, but its precision and F_1 vary with reference-label prevalence, class count, corruption structure, dataset size, and the classifier supplying probabilities. The strong AG News baseline is therefore evidence of cross-modal applicability, not a universal performance guarantee. The failure under concentrated 40% cyclic noise and the lower CIFAR-N precision identify conditions where review is more important than automated removal.

The project also changed the interpretation of "false positive." Half of the 30 blind-reviewed AG News cases favoured the model alternative even though no synthetic error had been injected. At the same time, Qwen's 46.7% order consistency shows that an LLM judge introduces its own uncertainty. A better future study would use multiple independent human annotators, a larger pre-registered sample, and adjudication of disagreements; it would also compare probability calibration before attributing differences solely to CL. Within the available CPU and annotation budget, the combination of exact artifacts, real human-noise data, controlled text data, and transparent judge evaluation provides a reproducible and appropriately qualified test of the paper's claims.

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A Supplementary Results

CIFAR-10N labels	Err.	F_1	MCC	AP	Clean acc.
Aggregate	466	.374	.337	.427	.692
Worst-case	2,023	.664	.382	.708	.470
Annotator 1	874	.480	.369	.564	.601
Annotator 2	908	.500	.390	.581	.604
Annotator 3	905	.494	.386	.560	.584

Table 11: Five CIFAR-10N human-label sets on the same 5,000 images and fixed folds.

Follow-up dataset	Candidates	Validated	Rate
MNIST	100	15	15.0%
CIFAR-10	275	54	19.6%
CIFAR-100	2,235	585	26.2%
Caltech-256	2,360	458	19.4%
ImageNet	5,440	2,916	53.6%
20 Newsgroups	93	82	88.2%
IMDB	1,310	725	55.3%
Total	11,813	4,835	-

Table 12: Exact reproduction of released human-validation counts from the follow-up test-set study. Rates are conditional on CL-flagged candidates.